



Qualification Pack

MACHINE OPERATOR - PLASTICS PROCESSING

QP Code: CPC/CAP/Q0101

Version: 1.0

NSQF Level: 3.5

CENTRAL INSTITUTE OF PETROCHEMICALS ENGINEERING & TECHNOLOGY || CIPET Head Office, T.V.K.
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CPC/CAP/Q0101: MACHINE OPERATOR - PLASTICS PROCESSING

Brief Job Description

The Machine operator handles the plastic granules (raw material), sets up and operates the plastic processing machines, finishes the product & stores in desired place.

Personal Attributes

Technical aptitude Safety-conscious Reliability and punctuality Organizational skills Problem-solving ability Teamwork and communication Quality-focused mindset Willingness to learn

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [DGT/VSQ/N0101: Employability Skills \(30 Hours\)](#)
2. [CPC/CAP/N0402: Basics of computer and data entry in MS OFFICE/office Open source suite Software](#)
3. [CPC/CAP/N0401: Basic health and safety practices at the workplace, 5S](#)
4. [CPC/CAP/N0105: Familiarized with various Plastics processing techniques & to assist the Operator in Extrusion & its Troubleshooting](#)
5. [CPC/CAP/N0104: Familiarized with various Plastics processing techniques & to assist the Operator in Blow Moulding & its Trouble shooting etc](#)
6. [CPC/CAP/N0103: Familiarized with various Plastics processing techniques & to assist the Operator in Injection Moulding & its Troubleshooting](#)
7. [CPC/CAP/N0102: To know about different plastic material](#)
8. [CPC/CAP/N0101: Basic concepts, job requirements & basic related process](#)

Qualification Pack (QP) Parameters

Sector	Chemical & PetroChemical
Sub-Sector	Petrochemicals
Occupation	Plastics Processing

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Country	India
NSQF Level	3.5
Credits	20
Aligned to NCO/ISCO/ISIC Code	NCO-2015/8142
Minimum Educational Qualification & Experience	11th Class with NA of experience OR 10th grade pass with 1.5 years of experience OR 8th grade pass with 4.5 years of experience OR Previous relevant Qualification of NSQF Level with 1.5 years of experience
Minimum Level of Education for Training in School	
Pre-Requisite License or Training	NA
Minimum Job Entry Age	18 Years
Last Reviewed On	NA
Next Review Date	25/05/2028
NSQC Approval Date	26/05/2025
Version	1.0
Reference code on NQR	QG-3.5-CP-04116-2025-V2-CIPET
NQR Version	2

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DGT/VSQ/N0101: Employability Skills (30 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

PC1. understand the significance of employability skills in meeting the job requirements

Constitutional values - Citizenship

To be competent, the user/individual on the job must be able to:

PC2. identify constitutional values, civic rights, duties, personal values and ethics and environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

PC3. explain 21st Century Skills such as Self-Awareness, Behavior Skills, Positive attitude, self-motivation, problem-solving, creative thinking, time management, social and cultural awareness, emotional awareness, continuous learning mindset etc.

Basic English Skills

To be competent, the user/individual on the job must be able to:

PC4. speak with others using some basic English phrases or sentences

Communication Skills

To be competent, the user/individual on the job must be able to:

PC5. follow good manners while communicating with others

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PC6. work with others in a team

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

PC7. communicate and behave appropriately with all genders and PwD

PC8. report any issues related to sexual harassment

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

PC9. use various financial products and services safely and securely

PC10. calculate income, expenses, savings etc.

PC11. approach the concerned authorities for any exploitation as per legal rights and laws

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

PC12. operate digital devices and use its features and applications securely and safely

PC13. use internet and social media platforms securely and safely

Entrepreneurship

To be competent, the user/individual on the job must be able to:

PC14. identify and assess opportunities for potential business

PC15. identify sources for arranging money and associated financial and legal challenges

Customer Service

To be competent, the user/individual on the job must be able to:

PC16. identify different types of customers

PC17. identify customer needs and address them appropriately

PC18. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC19. create a basic biodata

PC20. search for suitable jobs and apply

PC21. identify and register apprenticeship opportunities as per requirement

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use basic spoken English language

KU6. Do and dont of effective communication

KU7. inclusivity and its importance



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- KU8.** different types of disabilities and appropriate communication and behaviour towards PwD
- KU9.** different types of financial products and services
- KU10.** how to compute income and expenses
- KU11.** importance of maintaining safety and security in financial transactions
- KU12.** different legal rights and laws
- KU13.** how to operate digital devices and applications safely and securely
- KU14.** ways to identify business opportunities
- KU15.** types of customers and their needs
- KU16.** how to apply for a job and prepare for an interview
- KU17.** apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** communicate effectively using appropriate language
- GS2.** behave politely and appropriately with all
- GS3.** perform basic calculations
- GS4.** solve problems effectively
- GS5.** be careful and attentive at work
- GS6.** use time effectively
- GS7.** maintain hygiene and sanitisation to avoid infection

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. understand the significance of employability skills in meeting the job requirements	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC2. identify constitutional values, civic rights, duties, personal values and ethics and environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	1	3	-	-
PC3. explain 21st Century Skills such as Self-Awareness, Behavior Skills, Positive attitude, self-motivation, problem-solving, creative thinking, time management, social and cultural awareness, emotional awareness, continuous learning mindset etc.	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC4. speak with others using some basic English phrases or sentences	-	-	-	-
<i>Communication Skills</i>	1	1	-	-
PC5. follow good manners while communicating with others	-	-	-	-
PC6. work with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	1	-	-
PC7. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC8. report any issues related to sexual harassment	-	-	-	-
<i>Financial and Legal Literacy</i>	3	4	-	-
PC9. use various financial products and services safely and securely	-	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. calculate income, expenses, savings etc.	-	-	-	-
PC11. approach the concerned authorities for any exploitation as per legal rights and laws	-	-	-	-
<i>Essential Digital Skills</i>	4	6	-	-
PC12. operate digital devices and use its features and applications securely and safely	-	-	-	-
PC13. use internet and social media platforms securely and safely	-	-	-	-
<i>Entrepreneurship</i>	3	5	-	-
PC14. identify and assess opportunities for potential business	-	-	-	-
PC15. identify sources for arranging money and associated financial and legal challenges	-	-	-	-
<i>Customer Service</i>	2	2	-	-
PC16. identify different types of customers	-	-	-	-
PC17. identify customer needs and address them appropriately	-	-	-	-
PC18. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	1	3	-	-
PC19. create a basic biodata	-	-	-	-
PC20. search for suitable jobs and apply	-	-	-	-
PC21. identify and register apprenticeship opportunities as per requirement	-	-	-	-
NOS Total	20	30	-	-



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National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0101
NOS Name	Employability Skills (30 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	2
Credits	1
Version	1.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

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CPC/CAP/N0402: Basics of computer and data entry in MS OFFICE/office Open source suite Software

Description

This NOS unit is about the Data Entry Operation for Injection Moulding like entering, updating and maintain Job work related data the computer systems having MS Office software

Scope

The scope covers the following :

- This unit / task covers the Enter, update and maintain data in MS Office system

Elements and Performance Criteria

Enter, update and maintain data

To be competent, the user/individual on the job must be able to:

- PC1.** Fill and process mandated forms for receiving, processing, or tracking data, enter data from source documents (such as trial report, process sheet etc.) in to Computer application having MS OFFICE software.
- PC2.** Scan source documents in accordance with specific instructions.
- PC3.** Verify data entered with source documents, checks for compliance and corrects all typographical errors and missing or repeated data.
- PC4.** Maintain files of source documents or other information related to data entered.
- PC5.** Investigate and confirm data that is unclear before entering, generate reports of data entry, store completed work in designated locations and perform backup operations.
- PC6.** Update database information to reflect most current source information
- PC7.** Assist in the filing and storage of security and back up data files
- PC8.** Respond to requests for information and access relevant files

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** The data management applications/tools used by the company
- KU2.** Data entry protocol
- KU3.** Data integrity and security policies of the company
- KU4.** Approved methods for carrying document control and archiving
- KU5.** Basic understanding computers and its terminology
- KU6.** Work on different software needed for report writing including MS office suit or open source office

Generic Skills (GS)



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User/individual on the job needs to know how to:

- GS1.** Efficiently enter data into computer applications
- GS2.** Prepare legible reports
- GS3.** Read and understand manuals, SOPs, instructions, memos, reports, job cards etc.
- GS4.** Communicate effectively with the team members and supervisors
- GS5.** Apply basic logic to identify data errors
- GS6.** Pay attention to details
- GS7.** Plan assigned tasks within timeline and as per priority order specified
- GS8.** Identify process improvements

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Enter, update and maintain data</i>	8	14	-	-
PC1. Fill and process mandated forms for receiving, processing, or tracking data, enter data from source documents (such as trial report, process sheet etc.) in to Computer application having MS OFFICE software.	1	2	-	-
PC2. Scan source documents in accordance with specific instructions.	1	2	-	-
PC3. Verify data entered with source documents, checks for compliance and corrects all typographical errors and missing or repeated data.	1	2	-	-
PC4. Maintain files of source documents or other information related to data entered.	1	2	-	-
PC5. Investigate and confirm data that is unclear before entering, generate reports of data entry, store completed work in designated locations and perform backup operations.	1	2	-	-
PC6. Update database information to reflect most current source information	1	1	-	-
PC7. Assist in the filing and storage of security and back up data files	1	2	-	-
PC8. Respond to requests for information and access relevant files	1	1	-	-
NOS Total	8	14	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	CPC/CAP/N0402
NOS Name	Basics of computer and data entry in MS OFFICE/office Open source suite Software
Sector	Chemical & PetroChemical
Sub-Sector	
Occupation	Generic
NSQF Level	3.5
Credits	1
Version	1.0
Last Reviewed Date	26/05/2025
Next Review Date	25/05/2028
NSQC Clearance Date	26/05/2025

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CPC/CAP/N0401: Basic health and safety practices at the workplace, 5S

Description

This unit is about creating a Safe and Healthy work place, adhering to the safety guidelines in the working area, following practices which are not impacting the environment in a negative manner. This NOS is about ensuring all 5S activities both at the shop floor and the office area to facilitate increase in work productivity. It covers knowledge of fire safety, common first aid applications, safe practice and emergency applications.

Scope

The scope covers the following :

- The role holder will be responsible for
- Identifying and reporting of risks
- Creating and sustaining a safe, clean and environment friendly Work place.
- Ensure sorting, streamlining & organizing, storage and documentation, cleaning, standardization and sustenance across the plant and office premises of the organization

Elements and Performance Criteria

Health and safety

To be competent, the user/individual on the job must be able to:

- PC1.** Wear protective clothing/equipment for specific tasks and work conditions
- PC2.** Carry out safe working practices while dealing with hazards to ensure the safety of Self and others.
- PC3.** Apply good housekeeping practices at all times

Fire safety

To be competent, the user/individual on the job must be able to:

- PC4.** Use the various appropriate fire extinguishers on different types of fires correctly
- PC5.** Demonstrate rescue techniques applied during fire hazard, demonstrate good housekeeping in order to prevent fire hazards, demonstrate the correct use of a fire extinguisher.

Identify and report the risks identified

To be competent, the user/individual on the job must be able to:

- PC6.** Identify activities which can cause potential injury through sharp objects, burns, fall, electricity, gas leakages, radiation, poisonous fumes, chemicals, loud noise, and Identify areas in the plant which are potentially hazardous/ unhygienic in nature. Conduct regular checks with support of the maintenance team on machine health to identify potential hazards due to wear and tear of machine.
- PC7.** Inform the concerned authorities on the potential risks identified in the processes, workplace area/ layout, materials used etc, Inform the concerned authorities about machine breakdowns, damages which can potentially harm man/ machine during operations.
- PC8.** Create awareness amongst other by sharing information on the identified risks.

Ensure sorting

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To be competent, the user/individual on the job must be able to:

- PC9.** Follow the sorting process and check that the tools, fixtures & jigs that are lying on workstations are the ones in use and unnecessary items are not cluttering the workbenches or work surfaces.
- PC10.** Ensure segregation of waste in hazardous/ non Hazardous waste as per the sorting work instructions
- PC11.** Follow the technique of waste disposal and waste storage in the proper bins as per SOP
- PC12.** Segregate the items which are labeled as red tag items for the process area and keep them in the correct places
- PC13.** Sort the tools/ equipment/ fasteners/ spare parts as per specifications/ utility into proper trays, cabinets, lockers as mentioned in the 5S guidelines/ work instructions
- PC14.** Ensure that areas of material storage areas are not overflowing
- PC15.** Properly stack the various types of boxes and containers as per the size/ utility to avoid any fall of items/ breakage and also enable easy sorting when required
- PC16.** Return the extra material and tools to the designated sections and make sure that no additional material/ tool is lying near the work area
- PC17.** Follow the floor markings/ area markings used for demarcating the various sections in the plant as per the prescribed instructions and standards

Ensure proper documentation and storage(organizing, streamlining)

To be competent, the user/individual on the job must be able to:

- PC18.** Follow the proper labelling mechanism of instruments/ boxes/ containers and maintaining reference files/ documents with the codes and the lists
- PC19.** Check that the items in the respective areas have been identified as broken or damaged
- PC20.** Follow the given instructions and check for labelling of fluids, oils, lubricants, solvents, chemicals etc. and proper storage of the same. To avoid spillage, leakage, fire etc.
- PC21.** Make sure that all material and tools are stored in the designated places and in the manner indicated in the 5S instructions

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Relevant standards, procedures and policies related to Health, Safety and Environment followed in the company
- KU2.** Emergency handling procedures & hierarchy for escalation
- KU3.** Basic knowledge of Safety procedures(firefighting, first aid) within the organization
- KU4.** Basic knowledge of various types of PPEs and their usage
- KU5.** Basic knowledge of risks/hazards associated with each occupation in the organization
- KU6.** Knowledge of personal hygiene and how an individual can contribute towards creating a highly safe and clean working environment
- KU7.** Meaning of hazards and risks
- KU8.** Health and safety hazards commonly present in the work environment and related precautions

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- KU9.** Possible causes of risk, hazard or accident in the workplace and why risk and/or accidents are possible
- KU10.** Possible causes of risk and accident (due to oil leakage)
- KU11.** Safe working practices when working with tools and machines
- KU12.** Safe working practices while working at various hazardous sites
- KU13.** Where to find all the general health and safety equipment in the workplace
- KU14.** Various dangers associated with the use of electrical equipment
- KU15.** Preventative and remedial actions to be taken in the case of exposure to toxic materials
- KU16.** Importance of using protective clothing/equipment while working
- KU17.** Precautionary activities to prevent the fire accident
- KU18.** Various causes of fire
- KU19.** Techniques of using the different fire extinguishers
- KU20.** Different methods of extinguishing fire
- KU21.** Different materials used for extinguishing fire
- KU22.** Rescue techniques applied during a fire hazard
- KU23.** Various types of safety signs and what they mean
- KU24.** Appropriate basic first aid treatment relevant to the condition eg. shock, electrical shock, bleeding, breaks to bones, minor burns, resuscitation, poisoning, eye injuries
- KU25.** Content of written accident report
- KU26.** Potential injuries and ill health associated with incorrect manual handling
- KU27.** Safe lifting and carrying practices
- KU28.** Personal safety, health and dignity issues relating to the movement of a person by others
- KU29.** Potential impact to a person who is moved incorrectly
- KU30.** Have basic knowledge of 5S procedures
- KU31.** Know various types 5s practices followed in various areas
- KU32.** Understand the 5S checklists provided in the department/ team
- KU33.** Have skills to identify useful & non useful items
- KU34.** Have knowledge of labels , signs & colors used as indicators
- KU35.** Have knowledge on how to sort and store various types of tools, equipment, material etc.
- KU36.** Know , how to identify various types of waste products
- KU37.** Understand the impact of waste/ dirt/ dust/unwanted substances on the process/ environment/ machinery/ human body.
- KU38.** Have knowledge of best ways of cleaning & waste disposal

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Write basic level notes and observations
- GS2.** Read safety instructions put up across the plant premises

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- GS3.** Read safety precautions mentioned in equipment manuals and panels to understand the potential risks associated
- GS4.** Effectively communicate information to team members
- GS5.** Inform employees in the plant and concerned functions about events, incidents & potential risks observed related to Safety, Health and Environment.
- GS6.** Question operator/ supervisor in order to understand the safety related issues
- GS7.** Attentively listen with full attention and comprehend the information given by the speaker during safety drills and training programs
- GS8.** Plan and organize the work order and jobs received from the internal customers
- GS9.** Plan and organize the design documents received from internal customers
- GS10.** Organize all process/ equipment manuals so that sorting out information is Fast
- GS11.** Use common sense and make judgments during day to day basis
- GS12.** Use reasoning skills to identify and resolve basic problems
- GS13.** Use intuition to detect any potential problems which could arise during operations
- GS14.** Follow instructions and work on areas of improvement identified
- GS15.** Complete the assigned tasks with minimum supervision
- GS16.** Complete the job defined by the supervisor within the timelines and quality norms

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Health and safety</i>	1.5	6	-	-
PC1. Wear protective clothing/equipment for specific tasks and work conditions	0.5	2	-	-
PC2. Carry out safe working practices while dealing with hazards to ensure the safety of Self and others.	0.5	2	-	-
PC3. Apply good housekeeping practices at all times	0.5	2	-	-
<i>Fire safety</i>	1	4	-	-
PC4. Use the various appropriate fire extinguishers on different types of fires correctly	0.5	2	-	-
PC5. Demonstrate rescue techniques applied during fire hazard, demonstrate good housekeeping in order to prevent fire hazards, demonstrate the correct use of a fire extinguisher.	0.5	2	-	-
<i>Identify and report the risks identified</i>	1.5	6	-	-
PC6. Identify activities which can cause potential injury through sharp objects, burns, fall, electricity, gas leakages, radiation, poisonous fumes, chemicals, loud noise, and Identify areas in the plant which are potentially hazardous/ unhygienic in nature. Conduct regular checks with support of the maintenance team on machine health to identify potential hazards due to wear and tear of machine.	0.5	2	-	-
PC7. Inform the concerned authorities on the potential risks identified in the processes, workplace area/ layout, materials used etc, Inform the concerned authorities about machine breakdowns, damages which can potentially harm man/ machine during operations.	0.5	2	-	-
PC8. Create awareness amongst other by sharing information on the identified risks.	0.5	2	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Ensure sorting</i>	4	10	-	-
PC9. Follow the sorting process and check that the tools, fixtures & jigs that are lying on workstations are the ones in use and unnecessary items are not cluttering the workbenches or work surfaces.	0.5	2	-	-
PC10. Ensure segregation of waste in hazardous/ non Hazardous waste as per the sorting work instructions	0.5	2	-	-
PC11. Follow the technique of waste disposal and waste storage in the proper bins as per SOP	0.5	1	-	-
PC12. Segregate the items which are labeled as red tag items for the process area and keep them in the correct places	0.5	1	-	-
PC13. Sort the tools/ equipment/ fasteners/ spare parts as per specifications/ utility into proper trays, cabinets, lockers as mentioned in the 5S guidelines/ work instructions	0.5	1	-	-
PC14. Ensure that areas of material storage areas are not overflowing	0.25	0.5	-	-
PC15. Properly stack the various types of boxes and containers as per the size/ utility to avoid any fall of items/ breakage and also enable easy sorting when required	0.25	0.5	-	-
PC16. Return the extra material and tools to the designated sections and make sure that no additional material/ tool is lying near the work area	0.5	1	-	-
PC17. Follow the floor markings/ area markings used for demarcating the various sections in the plant as per the prescribed instructions and standards	0.5	1	-	-
<i>Ensure proper documentation and storage(organizing, streamlining)</i>	2	4	-	-
PC18. Follow the proper labelling mechanism of instruments/ boxes/ containers and maintaining reference files/ documents with the codes and the lists	0.5	1	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC19. Check that the items in the respective areas have been identified as broken or damaged	0.5	1	-	-
PC20. Follow the given instructions and check for labelling of fluids, oils, lubricants, solvents, chemicals etc. and proper storage of the same. To avoid spillage, leakage, fire etc.	0.5	1	-	-
PC21. Make sure that all material and tools are stored in the designated places and in the manner indicated in the 5S instructions	0.5	1	-	-
NOS Total	10	30	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	CPC/CAP/N0401
NOS Name	Basic health and safety practices at the workplace, 5S
Sector	Chemical & PetroChemical
Sub-Sector	
Occupation	Generic
NSQF Level	3.5
Credits	1
Version	1.0
Last Reviewed Date	26/05/2025
Next Review Date	25/05/2028
NSQC Clearance Date	26/05/2025

Qualification Pack

CPC/CAP/N0105: Familiarized with various Plastics processing techniques & to assist the Operator in Extrusion & its Troubleshooting

Description

This NOS unit is about operating the Extrusion Machine & its trouble shooting

Scope

The scope covers the following :

- The Machine Operator -Plastics Processing will be responsible for
- Pre-extrusion operation
- Extruder operation
- Trouble shooting
- Reporting & Documentation
- Sorting & Placing

Elements and Performance Criteria

Pre extrusion operation

To be competent, the user/individual on the job must be able to:

- PC1.** Planning work schedule in concurrence with Superior
- PC2.** Obtain and check the data on the job card and carry out functions in line with the responsibilities of job role
- PC3.** Ensure availability of data sheet, manual, work instructions
- PC4.** Check for power supply, oil level in gear box, water connections
- PC5.** Ensure availability & functioning of the tools ,materials & ancillary equipments I like Air Compressor, Cooling Tower, High Speed Mixer etc for the work
- PC6.** Setup the equipment & machineries as per the job requirement
- PC7.** Update and develop knowledge of the products to be produced
- PC8.** Planning for Minimum rejection & its safe reuse/disposal
- PC9.** Safety aspects of machine operation
- PC10.** Work in conformance to legal requirements, organizational policies and procedures

Extrusion

To be competent, the user/individual on the job must be able to:

- PC11.** Check material is available for production. Compounding / Color blending
- PC12.** Check the availability & readiness of ancillary equipments like air compressor, hopper loader, dehumidifier, Cooling towers etc
- PC13.** Load the material in the hopper
- PC14.** Set the parameters of the machine i.e temperatures, speeds etc
- PC15.** Check the temperature on the barrel with respect to set temperature
- PC16.** Conduct trial run to get extruded sample once machine is set

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- PC17.** Adjust parameters unless getting final product
- PC18.** Visual check of final product
- PC19.** Define accepted products and defective products as per approved plan
- PC20.** Corona treatment & printing, if required
- PC21.** Store the final product in specified area
- PC22.** Clean the machine & equipments at regular interval
- PC23.** Work in compliance with specified health and safety standards

Trouble Shooting

To be competent, the user/individual on the job must be able to:

- PC24.** Preventive maintenance of machines & ancillary equipments
- PC25.** Coordination with maintenance department for resolving breakdown maintenance in minimum possible time
- PC26.** Root cause analysis of extrusion defects
- PC27.** Analysis of data sheets available in department
- PC28.** Taking all corrective & preventive action

Reporting & documentation

To be competent, the user/individual on the job must be able to:

- PC29.** Reporting the problems caused by machines to superior, when not resolved by operator
- PC30.** Report defects in the moulds that one do not have the authority to repair
- PC31.** Report major processing defects beyond control of operator
- PC32.** Keeping records of machine log book, data sheet of machine parameter
- PC33.** Documents related to incoming & outgoing material

Achieve productivity, quality and safety standards as per company's norms

To be competent, the user/individual on the job must be able to:

- PC34.** Meet targets & goals for production
- PC35.** Minimise defects in final product
- PC36.** Follow quality system to get better product
- PC37.** Keep work area clean & systematic
- PC38.** Comply to safety & health guidelines & rules

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Company policies on personnel management
- KU2.** Company code of conduct & policy
- KU3.** Importance of individual role in the work flow
- KU4.** Organisation culture
- KU5.** Company reporting structure
- KU6.** Functional process like store management, procurement, quality management

Qualification Pack

- KU7.** Understand different types of extrusion grade plastic material
- KU8.** Understanding properties of above plastic material
- KU9.** Knowledge of Extrusion Machine operation
- KU10.** Machine start up procedure
- KU11.** Principle of Extrusion, Blown film , Pipe/ Profile Extrusion
- KU12.** Screw, L/D ratio, different types of screws
- KU13.** Single Screw & Twin Screw Extrusion
- KU14.** Parameter setting of Extrusion Machine Temperature, Speed & Time
- KU15.** Die setting, breaker plate, screen pack , back pressure
- KU16.** Sizing methods, Calibrating unit / collapsing unit, Hall off/take off & winding unit, Cutting & sealing unit for films
- KU17.** Extruder Unit, Gear box, motor, drive
- KU18.** Extrusion rate calculation
- KU19.** Monitoring of parameters for production of quality components
- KU20.** Quality Control & testing of plastic product
- KU21.** Minimisation of rejection & reuse
- KU22.** Shut down procedure

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Prepare document related to processing parameter, other technical records like machine log sheets, job card etc.
- GS2.** Write information documents to internal departments/ internal teams
- GS3.** Compilation of production records
- GS4.** Read & interpret machine parameters
- GS5.** Read equipment manuals and process documents
- GS6.** Read instructions like safety instructions , symbols while using the equipment in the plant area
- GS7.** Communicate orally any instructions related to work with superiors & co workers with clarity
- GS8.** Listen actively
- GS9.** Follow company protocol for communication
- GS10.** Make proper decisions pertaining to the work
- GS11.** Identification of problem
- GS12.** Finding the resource to resolve the problem
- GS13.** Consult superiors in case of any assistance
- GS14.** Plan, fix up priorities for work operations as per job requirements
- GS15.** organize and analyze information relevant to work
- GS16.** Basic concepts of shop-floor work productivity including material management waste reduction etc.



Qualification Pack

- GS17.** Undertake and express new ideas and initiatives to others
- GS18.** Modify work plan to overcome unforeseen difficulties or developments that occur as work progresses
- GS19.** Participate in improvement procedures including process, quality etc
- GS20.** Apply, analyze, and evaluate the information gathered from observation, experience, reasoning, or communication, as a guide to thought and action
- GS21.** Exhibit good team work with all
- GS22.** Consult superiors or fellow workers in case of any assistance
- GS23.** Maintains good inter personal relations

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Pre extrusion operation</i>	15	20	-	-
PC1. Planning work schedule in concurrence with Superior	1.5	2	-	-
PC2. Obtain and check the data on the job card and carry out functions in line with the responsibilities of job role	1.5	2	-	-
PC3. Ensure availability of data sheet, manual, work instructions	1.5	2	-	-
PC4. Check for power supply, oil level in gear box, water connections	1.5	2	-	-
PC5. Ensure availability & functioning of the tools ,materials & ancillary equipments I like Air Compressor, Cooling Tower, High Speed Mixer etc for the work	1.5	2	-	-
PC6. Setup the equipment & machineries as per the job requirement	1.5	2	-	-
PC7. Update and develop knowledge of the products to be produced	1.5	2	-	-
PC8. Planning for Minimum rejection & its safe reuse/disposal	1.5	2	-	-
PC9. Safety aspects of machine operation	1.5	2	-	-
PC10. Work in conformance to legal requirements, organizational policies and procedures	1.5	2	-	-
<i>Extrusion</i>	18	42	-	-
PC11. Check material is available for production. Compounding / Color blending	1.5	2	-	-
PC12. Check the availability & readiness of ancillary equipments like air compressor, hopper loader, dehumidifier, Cooling towers etc	1.5	3.5	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC13. Load the material in the hopper	1.5	3.5	-	-
PC14. Set the parameters of the machine i.e temperatures, speeds etc	1.5	3.5	-	-
PC15. Check the temperature on the barrel with respect to set temperature	1.5	3.5	-	-
PC16. Conduct trial run to get extruded sample once machine is set	1.5	3.5	-	-
PC17. Adjust parameters unless getting final product	1.5	3.5	-	-
PC18. Visual check of final product	1.5	3.5	-	-
PC19. Define accepted products and defective products as per approved plan	1.5	3.5	-	-
PC20. Corona treatment & printing, if required	1.5	3.5	-	-
PC21. Store the final product in specified area	1.5	3.5	-	-
PC22. Clean the machine & equipments at regular interval	0.5	2	-	-
PC23. Work in compliance with specified health and safety standards	1	3	-	-
<i>Trouble Shooting</i>	6.5	14.5	-	-
PC24. Preventive maintenance of machines & ancillary equipments	1	2	-	-
PC25. Coordination with maintenance department for resolving breakdown maintenance in minimum possible time	1	2	-	-
PC26. Root cause analysis of extrusion defects	1.5	3.5	-	-
PC27. Analysis of data sheets available in department	1.5	3.5	-	-
PC28. Taking all corrective & preventive action	1.5	3.5	-	-
<i>Reporting & documentation</i>	5.5	13.5	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC29. Reporting the problems caused by machines to superior, when not resolved by operator	1.5	3.5	-	-
PC30. Report defects in the moulds that one do not have the authority to repair	1	3.5	-	-
PC31. Report major processing defects beyond control of operator	1	2.5	-	-
PC32. Keeping records of machine log book, data sheet of machine parameter	1	2	-	-
PC33. Documents related to incoming & outgoing material	1	2	-	-
<i>Achieve productivity, quality and safety standards as per company's norms</i>	5	10	-	-
PC34. Meet targets & goals for production	1	2	-	-
PC35. Minimise defects in final product	1	2	-	-
PC36. Follow quality system to get better product	1	2	-	-
PC37. Keep work area clean & systematic	1	2	-	-
PC38. Comply to safety & health guidelines & rules	1	2	-	-
NOS Total	50	100	-	-



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	CPC/CAP/N0105
NOS Name	Familiarized with various Plastics processing techniques & to assist the Operator in Extrusion & its Troubleshooting
Sector	Chemical & PetroChemical
Sub-Sector	
Occupation	Plastics Processing
NSQF Level	3.5
Credits	4
Version	1.0
Last Reviewed Date	26/05/2025
Next Review Date	25/05/2028
NSQC Clearance Date	26/05/2025

Qualification Pack

CPC/CAP/N0104: Familiarized with various Plastics processing techniques & to assist the Operator in Blow Moulding & its Trouble shooting etc

Description

This NOS is about understanding the Blow molding process, mould / materials used, troubleshooting of process

Scope

The scope covers the following :

- The Machine Operator Plastics Processing will be responsible for
- Understanding the Process & process requirements
- Arranging the required raw material and tools for the process
- Troubleshooting & housekeeping
- Reporting & Documentation
- Achieve productivity, quality and safety standards as per company norms

Elements and Performance Criteria

Understanding the Process & process requirements

To be competent, the user/individual on the job must be able to:

- PC1.** understand the process, their types, operations involved
- PC2.** Discuss the work requirements for the process and with the supervisor
- PC3.** Refer all components / process related documents to understand dimensions and properties of the required work output
- PC4.** Understand the process requirements in terms of tools / mould / die required, temperature of the heater according to plastics material being used, Hydraulic / pneumatic pressure / rotating speed of the screw, Parison formation, Parison Programming, Blowing time etc. as mentioned in the Work Instruction / SOP / Control Diagrams
- PC5.** Clearly understanding the do's and don'ts of the blow molding process as defined in SOPs / Work Instructions or as defined by supervisors

Arranging the required raw material and tools for the process

To be competent, the user/individual on the job must be able to:

- PC6.** Understand the conversion procedure and process to be adopted for completing the work order from the supervisor by referring the Work Instruction document / SOP manual
- PC7.** Set the various parameters like temperature of the heaters, temperature of chiller / cooling tower, hydraulic / air pressure, rotating speed of the screw, screw pressure, regulating current, flow of coolant / water etc. before starting the process as per the parameters mentioned in the Work Instructions / SOP
- PC8.** Understand the raw material like plastics granules, bonding additives etc. required for production
- PC9.** Ensure that the required material with enough stock is available before starting the process

Qualification Pack

PC10. Understand the type of Mould / Die required to complete the conversion operation and ensure that the same is available for moulding operations

PC11. Ensure the availability of spare parts for continuous operation of machine

Troubleshooting & housekeeping

To be competent, the user/individual on the job must be able to:

PC12. Understand the troubleshooting of the blow molding process. Knows the quality defects observed in blow molding, their causes and remedies

PC13. Set the parameters to ensure manufacturing of good product

PC14. Ensure that mould / Die are cleaned properly & no foreign material is trapped in parts of mould/die

PC15. Ensure cleaning of the other moulding machine tools, auxiliaries (if any)

PC16. Ensure cleaning of the area around the machine for any oil, grease, water etc

Reporting & documentation

To be competent, the user/individual on the job must be able to:

PC17. Reporting the problems caused by machines to superior, when not resolved by operator

PC18. Report defects in the moulds that one do not have the authority to repair

PC19. Report major processing defects beyond control of operator

PC20. Keeping records of machine log book, data sheet of machine parameter

PC21. Documents related to incoming & outgoing material

Achieve productivity, quality and safety standards as per company's norms

To be competent, the user/individual on the job must be able to:

PC22. Meet targets & goals for production

PC23. Minimise defects in final product

PC24. Follow quality system to get better product

PC25. Keep work area clean & systematic

PC26. Comply to safety & health guidelines & rules

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. company policies on personnel management

KU2. company code of conduct & policy

KU3. importance of individual role in the work flow

KU4. organization culture

KU5. company reporting structure

KU6. Functional process like store management, procurement, quality management

KU7. understanding of blow molding processing technique, working principle, operating procedure etc

Qualification Pack

- KU8.** different parameters to process like heater temperature, hydraulic pressure/ air pressure/ vacuum pressure, rotating speed of the screw, operating current and voltage, injection time, refilling time, blowing time etc. and the impact of these parameters on the process output
- KU9.** various types of plastics materials like LDPE, HDPE, PP etc. being use for the process and the additives / master batches to be used
- KU10.** Processing behavior of various plastic raw materials being used
- KU11.** Safe storage of raw materials, their mixing, blending, etc
- KU12.** Types of moulds & dies used their operation
- KU13.** different types of measuring instruments like vernier callipers, micrometres Etc. for geometry and dimension measurement of the product
- KU14.** different types of tools to trim the plastic product
- KU15.** hazards and safety aspects involved in different processing techniques

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** prepare document related to processing parameter, other technical records like machine log sheets, job card etc
- GS2.** prepare draft drawings for the final output product
- GS3.** write information documents to internal departments/ internal teams
- GS4.** read & interpret machine parameters
- GS5.** read and interpret engineering drawing and sketches
- GS6.** read equipment manuals and process documents
- GS7.** read instructions like safety instructions , symbols while using the equipment in the plant area
- GS8.** Communicate orally any instructions related to work with superiors & co workers with clarity
- GS9.** Listen actively
- GS10.** Follow company protocol for communication
- GS11.** make proper decisions pertaining to the work
- GS12.** Identification of problem
- GS13.** Finding the resource to resolve the problem
- GS14.** consult superiors in case of any assistance
- GS15.** plan, fix up priorities for work operations as per job requirements
- GS16.** organize and analyze information relevant to work
- GS17.** basic concepts of shop-floor work productivity including material management waste reduction etc.
- GS18.** undertake and express new ideas and initiatives to others
- GS19.** modify work plan to overcome unforeseen difficulties or developments that occur as work progresses
- GS20.** participate in improvement procedures including process, quality etc



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- GS21.** apply, analyze, and evaluate the information gathered from observation, experience, reasoning, or communication, as a guide to thought and action
- GS22.** exhibit good team work with all
- GS23.** Consult superiors or fellow workers in case of any assistance
- GS24.** Maintains good inter personal relations

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Understanding the Process & process requirements</i>	5	10	-	-
PC1. understand the process, their types, operations involved	1	2	-	-
PC2. Discuss the work requirements for the process and with the supervisor	1	2	-	-
PC3. Refer all components / process related documents to understand dimensions and properties of the required work output	1	2	-	-
PC4. Understand the process requirements in terms of tools / mould / die required, temperature of the heater according to plastics material being used, Hydraulic / pneumatic pressure / rotating speed of the screw, Parison formation, Parison Programming, Blowing time etc. as mentioned in the Work Instruction / SOP / Control Diagrams	1	2	-	-
PC5. Clearly understanding the do's and don'ts of the blow molding process as defined in SOPs / Work Instructions or as defined by supervisors	1	2	-	-
<i>Arranging the required raw material and tools for the process</i>	6	12	-	-
PC6. Understand the conversion procedure and process to be adopted for completing the work order from the supervisor by referring the Work Instruction document / SOP manual	1	2	-	-
PC7. Set the various parameters like temperature of the heaters, temperature of chiller / cooling tower, hydraulic / air pressure, rotating speed of the screw, screw pressure, regulating current, flow of coolant / water etc. before starting the process as per the parameters mentioned in the Work Instructions / SOP	1	2	-	-
PC8. Understand the raw material like plastics granules, bonding additives etc. required for production	1	2	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC9. Ensure that the required material with enough stock is available before starting the process	1	2	-	-
PC10. Understand the type of Mould / Die required to complete the conversion operation and ensure that the same is available for moulding operations	1	2	-	-
PC11. Ensure the availability of spare parts for continuous operation of machine	1	2	-	-
<i>Troubleshooting & housekeeping</i>	5	16	-	-
PC12. Understand the troubleshooting of the blow molding process. Knows the quality defects observed in blow molding, their causes and remedies	1	2	-	-
PC13. Set the parameters to ensure manufacturing of good product	1	3	-	-
PC14. Ensure that mould / Die are cleaned properly & no foreign material is trapped in parts of mould/die	1	3	-	-
PC15. Ensure cleaning of the other moulding machine tools, auxiliaries (if any)	1	4	-	-
PC16. Ensure cleaning of the area around the machine for any oil, grease, water etc	1	4	-	-
<i>Reporting & documentation</i>	6.5	18	-	-
PC17. Reporting the problems caused by machines to superior, when not resolved by operator	1	4	-	-
PC18. Report defects in the moulds that one do not have the authority to repair	1	4	-	-
PC19. Report major processing defects beyond control of operator	1.5	4	-	-
PC20. Keeping records of machine log book, data sheet of machine parameter	1.5	3	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC21. Documents related to incoming & outgoing material	1.5	3	-	-
<i>Achieve productivity, quality and safety standards as per company's norms</i>	7.5	14	-	-
PC22. Meet targets & goals for production	1.5	3	-	-
PC23. Minimise defects in final product	1.5	3	-	-
PC24. Follow quality system to get better product	1.5	3	-	-
PC25. Keep work area clean & systematic	1.5	3	-	-
PC26. Comply to safety & health guidelines & rules	1.5	2	-	-
NOS Total	30	70	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	CPC/CAP/N0104
NOS Name	Familiarized with various Plastics processing techniques & to assist the Operator in Blow Moulding & its Trouble shooting etc
Sector	Chemical & PetroChemical
Sub-Sector	
Occupation	Plastics Processing, Plastics Processing
NSQF Level	3.5
Credits	4
Version	1.0
Last Reviewed Date	26/05/2025
Next Review Date	25/05/2028
NSQC Clearance Date	26/05/2025

Qualification Pack

CPC/CAP/N0103: Familiarized with various Plastics processing techniques & to assist the Operator in Injection Moulding & its Troubleshooting

Description

This NOS is about operating the Injection Moulding Machine & its trouble shooting

Scope

The scope covers the following :

- The Machine Operator Plastics Processing will be responsible for
- Pre moulding operation
- Moulding Operation
- Trouble shooting
- Reporting & Documentation
- Achieve productivity, quality and safety standards as per company norms

Elements and Performance Criteria

Pre moulding operation

To be competent, the user/individual on the job must be able to:

- PC1.** Planning work schedule in concurrence with Superior
- PC2.** Obtain and check the data on the job card and carry out functions in line with the responsibilities of job role
- PC3.** Ensure availability of data sheet, manual, work instructions
- PC4.** for power supply, hydraulic oil level, water connections
- PC5.** Ensure availability of the tools ,materials & ancillary equipments for the work
- PC6.** Setup the equipment & machineries as per the job requirement
- PC7.** Update and develop knowledge of the products
- PC8.** Planning for Minimum wastage & its safe disposal
- PC9.** Work in conformance to legal requirements, organizational policies and procedures

Moulding Operation

To be competent, the user/individual on the job must be able to:

- PC10.** Ensure that the mould is ready & having no problem in dry run
- PC11.** Check material is available for production. If required arrange for pre drying
- PC12.** Check the availability & readiness of ancillary equipments like chiller, mould Temperature controller, hopper loader, Cooling towers etc
- PC13.** Load the material and pigment (if required) in the hopper
- PC14.** Set the parameters of the machine i.e temperature, pressure, speed etc
- PC15.** Check the temperature on the barrel with respect to set temperature
- PC16.** Conduct trial run to get sample piece once machine is set

Qualification Pack

- PC17.** Adjust parameters unless getting final product
- PC18.** Visual check of final product
- PC19.** Define accepted products and defective products as per approved plan
- PC20.** Carry out post molding operation during the cycle time run such as. trimming, apply protective tapes, putting labels on each product for identification
- PC21.** Store the final product in specified area
- PC22.** Clean the machine & equipments at regular interval
- PC23.** Work in compliance with specified health and safety standards

Trouble Shooting

To be competent, the user/individual on the job must be able to:

- PC24.** Preventive maintenance of machines & ancillary equipments
- PC25.** Coordination with maintenance department for resolving breakdown maintenance in minimum possible time.
- PC26.** Root cause analysis of moulding defects
- PC27.** Analysis of data sheets available in department
- PC28.** Taking all corrective & preventive action

Reporting & documentation

To be competent, the user/individual on the job must be able to:

- PC29.** Reporting the problems caused by machines to superior, when not resolved by operator
- PC30.** Report defects in the moulds that one do not have the authority to repair
- PC31.** Report major processing defects beyond control of operator
- PC32.** Keeping records of machine log book, data sheet of machine parameter
- PC33.** Documents related to incoming & outgoing material

Achieve productivity, quality and safety standards as per company's norms

To be competent, the user/individual on the job must be able to:

- PC34.** Meet targets & goals for production
- PC35.** Minimise defects in final product
- PC36.** Follow quality system to get better product
- PC37.** Keep work area clean & systematic
- PC38.** Comply to safety & health guidelines & rules

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** company policies on personnel management
- KU2.** company code of conduct & policy
- KU3.** importance of individual role in the work flow
- KU4.** organisation culture
- KU5.** company reporting structure
- KU6.** functional process like store management, procurement, quality management

Qualification Pack

- KU7.** Understand different types of plastic material
- KU8.** Understanding properties of plastic material
- KU9.** Knowledge of Semi-Automatic & Fully-Automatic operation of machines
- KU10.** Machine start up procedure
- KU11.** Principle of Injection Moulding
- KU12.** Parameter setting of injection moulding Machine Temperature, Pressure, Time
- KU13.** Clamping system Hydraulic & Toggle
- KU14.** Mould loading & unloading procedure
- KU15.** Calculation of tonnage, importance of mould dimensions, mould day-light
- KU16.** Injection Unit, shot weight setting, Calculation of plasticizing capacity of Machine , types of nozzles, ring plunger set
- KU17.** Monitoring of parameters for production of quality components
- KU18.** Post moulding operation like finishing, deflashing
- KU19.** Quality Control & testing of plastic product
- KU20.** Minimisation of rejection & reuse of feed system
- KU21.** shut down procedure

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** prepare document related to processing parameter, other technical records like machine log sheets, job card etc.
- GS2.** write information documents to internal departments/ internal teams
- GS3.** compilation of production records
- GS4.** read & interpret machine parameters
- GS5.** read equipment manuals and process documents
- GS6.** read instructions like safety instructions , symbols while using the equipment in the plant area
- GS7.** Communicate orally any instructions related to work with superiors & co workers with clarity
- GS8.** Listen actively
- GS9.** Follow company protocol for communication
- GS10.** make proper decisions pertaining to the work
- GS11.** Identification of problem
- GS12.** Finding the resource to resolve the problem
- GS13.** consult superiors in case of any assistance
- GS14.** plan, fix up priorities for work operations as per job requirements
- GS15.** organize and analyze information relevant to work
- GS16.** basic concepts of shop-floor work productivity including material management waste reduction etc
- GS17.** undertake and express new ideas and initiatives to others



Qualification Pack

- GS18.** modify work plan to overcome unforeseen difficulties or developments that occur as work progresses
- GS19.** participate in improvement procedures including process, quality etc
- GS20.** apply, analyze, and evaluate the information gathered from observation, experience, reasoning, or communication, as a guide to thought and action
- GS21.** exhibit good team work with all
- GS22.** Consult superiors or fellow workers in case of any assistance
- GS23.** Maintains good inter personal relations

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Pre moulding operation</i>	9	19	-	-
PC1. Planning work schedule in concurrence with Superior	1	2	-	-
PC2. Obtain and check the data on the job card and carry out functions in line with the responsibilities of job role	1	2	-	-
PC3. Ensure availability of data sheet, manual, work instructions	1	1	-	-
PC4. for power supply, hydraulic oil level, water connections	1	2	-	-
PC5. Ensure availability of the tools ,materials & ancillary equipments for the work	1	2	-	-
PC6. Setup the equipment & machineries as per the job requirement	1	2	-	-
PC7. Update and develop knowledge of the products	1	2	-	-
PC8. Planning for Minimum wastage & its safe disposal	1	2	-	-
PC9. Work in conformance to legal requirements, organizational policies and procedures	1	4	-	-
<i>Moulding Operation</i>	26	49	-	-
PC10. Ensure that the mould is ready & having no problem in dry run	1	4	-	-
PC11. Check material is available for production. If required arrange for pre drying	2	4	-	-
PC12. Check the availability & readiness of ancillary equipments like chiller, mould Temperature controller, hopper loader, Cooling towers etc	2	4	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC13. Load the material and pigment (if required) in the hopper	2	4	-	-
PC14. Set the parameters of the machine i.e temperature, pressure, speed etc	2	4	-	-
PC15. Check the temperature on the barrel with respect to set temperature	2	4	-	-
PC16. Conduct trial run to get sample piece once machine is set	2	3	-	-
PC17. Adjust parameters unless getting final product	2	3	-	-
PC18. Visual check of final product	2	3	-	-
PC19. Define accepted products and defective products as per approved plan	2	3	-	-
PC20. Carry out post molding operation during the cycle time run such as. trimming, apply protective tapes, putting labels on each product for identification	2	3	-	-
PC21. Store the final product in specified area	2	3	-	-
PC22. Clean the machine & equipments at regular interval	2	4	-	-
PC23. Work in compliance with specified health and safety standards	1	3	-	-
Trouble Shooting	5	12	-	-
PC24. Preventive maintenance of machines & ancillary equipments	1	3	-	-
PC25. Coordination with maintenance department for resolving breakdown maintenance in minimum possible time.	1	3	-	-
PC26. Root cause analysis of moulding defects	1	2	-	-
PC27. Analysis of data sheets available in department	1	2	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC28. Taking all corrective & preventive action	1	2	-	-
<i>Reporting & documentation</i>	5	10	-	-
PC29. Reporting the problems caused by machines to superior, when not resolved by operator	1	2	-	-
PC30. Report defects in the moulds that one do not have the authority to repair	1	2	-	-
PC31. Report major processing defects beyond control of operator	1	2	-	-
PC32. Keeping records of machine log book, data sheet of machine parameter	1	2	-	-
PC33. Documents related to incoming & outgoing material	1	2	-	-
<i>Achieve productivity, quality and safety standards as per company's norms</i>	5	10	-	-
PC34. Meet targets & goals for production	1	2	-	-
PC35. Minimise defects in final product	1	2	-	-
PC36. Follow quality system to get better product	1	2	-	-
PC37. Keep work area clean & systematic	1	2	-	-
PC38. Comply to safety & health guidelines & rules	1	2	-	-
NOS Total	50	100	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	CPC/CAP/N0103
NOS Name	Familiarized with various Plastics processing techniques & to assist the Operator in Injection Moulding & its Troubleshooting
Sector	Chemical & PetroChemical
Sub-Sector	
Occupation	Plastics Processing, Plastics Processing
NSQF Level	3.5
Credits	5
Version	1.0
Last Reviewed Date	26/05/2025
Next Review Date	25/05/2028
NSQC Clearance Date	26/05/2025

Qualification Pack

CPC/CAP/N0102: To know about different plastic material

Description

This NOS is about understanding the different types of plastics materials being used in the industry, their basic knowhow, properties, etc.

Scope

The scope covers the following :

- understanding the various types of Plastics materials, basic knowhow of the processing behavior of Plastics materials and maintaining the raw material for the process

Elements and Performance Criteria

understanding the various types of Plastics materials

To be competent, the user/individual on the job must be able to:

- PC1.** Discuss about the type of raw material being used in the industry & for work Order required for the process and with the supervisor
- PC2.** Refer all material related documents to understand properties of the required work output and able to identify the material
- PC3.** Understand the process requirements for the Plastics material in terms of temperature of the heater, rotating speed of the Screw, pressure, injection as mentioned in the Work Instruction / SOP / Control Diagrams

basic know how of the processing behavior of Plastics materials

To be competent, the user/individual on the job must be able to:

- PC4.** Understand the melting temperature, processing temperature etc. for plastic raw material
- PC5.** Understand the processing characteristics of the plastics material being used for conversion procedure and process to be adopted for completing the work order from the supervisor by referring the Work Instruction document / SOP manual

maintaining the raw material for the process

To be competent, the user/individual on the job must be able to:

- PC6.** Ensure that the required material is available before starting the process
- PC7.** Ensure that the plastics material is blended with requisite additives

cleaning the material spillage around machine

To be competent, the user/individual on the job must be able to:

- PC8.** Ensure that machine / mould / Die are cleaned properly & no foreign material is entrapped in parts of the machine / mould / die.
- PC9.** Ensure cleaning of the materials spilled around the machine
- PC10.** Ensure cleaning of the area around the machine for any oil, grease, water etc

Knowledge and Understanding (KU)

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The individual on the job needs to know and understand:

- KU1.** company policies on personnel management
- KU2.** company code of conduct & policy
- KU3.** importance of individual role in the work flow
- KU4.** organization culture
- KU5.** company reporting structure
- KU6.** Functional process like store management, procurement, quality management
- KU7.** understanding different of plastic materials
- KU8.** various types of plastics like thermoplastics/ thermosetting plastics and the additives to be used
- KU9.** properties / characteristics of various plastic materials
- KU10.** Processing behavior of various plastic raw materials
- KU11.** Safe storage of raw materials
- KU12.** hazards and safety aspects involved with different processing techniques

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** prepare document related to raw material used, stock position, other technical records like machine log sheets, job card etc.
- GS2.** write information documents to internal departments/ internal teams
- GS3.** read & interpret material data sheet
- GS4.** read & interpret machine parameters
- GS5.** read instructions like safety instructions , symbols being used in the plant area
- GS6.** Communicate orally any instructions related to work with superiors & co- workers with clarity
- GS7.** Listen carefully & follow company protocol for communication
- GS8.** Identification of problem
- GS9.** make proper decisions pertaining to the work
- GS10.** consult superiors in case of any assistance
- GS11.** fix up priorities for work operations as per job requirements
- GS12.** organize and analyze information relevant to work
- GS13.** basic concepts of shop-floor work productivity including material management, waste reduction etc
- GS14.** undertake and express new ideas and initiatives to others
- GS15.** modify work plan to overcome unforeseen difficulties or developments that occur as work progresses
- GS16.** participate in improvement of procedures including process, quality etc
- GS17.** apply, analyze, and evaluate the information gathered from observation, experience, reasoning, or communication, as a guide to thought and action



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- GS18.** exhibit good team work with all
- GS19.** Maintains good inter personal relations
- GS20.** Consult superiors or fellow workers in case of any assistance

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>understanding the various types of Plastics materials</i>	6	8	-	-
PC1. Discuss about the type of raw material being used in the industry & for work Order required for the process and with the supervisor	2	2	-	-
PC2. Refer all material related documents to understand properties of the required work output and able to identify the material	2	3	-	-
PC3. Understand the process requirements for the Plastics material in terms of temperature of the heater, rotating speed of the Screw, pressure, injection as mentioned in the Work Instruction / SOP / Control Diagrams	2	3	-	-
<i>basic know how of the processing behavior of Plastics materials</i>	4	6	-	-
PC4. Understand the melting temperature, processing temperature etc. for plastic raw material	2	3	-	-
PC5. Understand the processing characteristics of the plastics material being used for conversion procedure and process to be adopted for completing the work order from the supervisor by referring the Work Instruction document / SOP manual	2	3	-	-
<i>maintaining the raw material for the process</i>	4	6	-	-
PC6. Ensure that the required material is available before starting the process	2	3	-	-
PC7. Ensure that the plastics material is blended with requisite additives	2	3	-	-
<i>cleaning the material spillage around machine</i>	6	10	-	-
PC8. Ensure that machine / mould / Die are cleaned properly & no foreign material is entrapped in parts of the machine / mould / die.	2	3	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC9. Ensure cleaning of the materials spilled around the machine	2	4	-	-
PC10. Ensure cleaning of the area around the machine for any oil, grease, water etc	2	3	-	-
NOS Total	20	30	-	-



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	CPC/CAP/N0102
NOS Name	To know about different plastic material
Sector	Chemical & PetroChemical
Sub-Sector	
Occupation	Plastics Processing
NSQF Level	3.5
Credits	2
Version	1.0
Last Reviewed Date	26/05/2025
Next Review Date	25/05/2028
NSQC Clearance Date	26/05/2025

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CPC/CAP/N0101: Basic concepts, job requirements & basic related process

Description

This module covers the basic concepts of occupational safety, types and usage of personal protective equipment, emergency response procedures, identification and control of workplace hazards, and the importance of maintaining hygiene and a clean work environment in the plastic processing industry.

Scope

The scope covers the following :

- To equip learners with foundational knowledge and practical skills related to workplace safety, use of PPE, hazard identification, and hygiene practices in plastic processing environments.

Elements and Performance Criteria

Understand the work order and the process requirements

To be competent, the user/individual on the job must be able to:

- PC1.** Discuss the work order (work output) required from the process and with the supervisor
- PC2.** Refer all components / process related documents to understand dimensions and properties of the required work output
- PC3.** Understand the process requirements in terms of temperature of the heater, hydraulic pressure/ air pressure/ vacuum pressure, rotating speed of the screw pressure, injection time, refilling time, blowing time etc. as mentioned in the Work Instruction/ SOP/ Control Diagrams
- PC4.** Clearly understanding the does and don'ts of the manufacturing process as defined in SOPs/ Work Instructions or defined by supervisors

Arrange for the material to be processed and apparatus required for the same

To be competent, the user/individual on the job must be able to:

- PC5.** Understand the conversion procedure and process to be adopted for completing the work order from the supervisor by referring the Work Instruction document/ SOP manual
- PC6.** Set the various parameters like temperature of the heaters, hydraulic pressure/air pressure/ vacuum pressure, rotating speed of the screw, screw pressure, regulating current, flow of coolant/ water etc. before starting the process as per the parameters are mentioned in the Work Instructions/ SOP manual
- PC7.** Understand the raw material like plastics granules, bonding additives etc. required for executing the activity
- PC8.** Ensure that the required material is available before starting the process
- PC9.** Understand the type of Mould /Die required for executing the required conversion operation and ensure that the same is available for moulding operations
- PC10.** Ensure the availability of spare parts for continuous operation of machine

Clean the apparatus and the components before executing the process

To be competent, the user/individual on the job must be able to:

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PC11. Ensure that mould / Die are cleaned properly & no foreign material is entrapped in parts of mould/die

PC12. Ensure cleaning of the other moulding machine tools, auxiliaries(if any)

PC13. Ensure cleaning of the area around the machine for any oil, grease, water etc

Escalations of queries on the given job

To be competent, the user/individual on the job must be able to:

PC14. Consult with superiors in case of any doubt/clarification

PC15. Self-confidence after resolving the queries to complete the task

PC16. Report completion of work to superiors

Interaction with other concern department

To be competent, the user/individual on the job must be able to:

PC17. Good interpersonal relations with superiors & fellow operators

PC18. Disciplined behavior in work place

PC19. Good coordination with other department person for getting their support for work

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. Company's policies on personnel management

KU2. company code of conduct & policy

KU3. importance of individual role in the work flow

KU4. Organisation culture

KU5. Company reporting structure

KU6. Functional process like store management, procurement, quality management

KU7. Understanding different types of plastic processing techniques

KU8. Different parameters pertinent to process like heater temperature, hydraulic pressure/ air pressure/ vacuum pressure, rotating speed of the screw, operating current and voltage, injection time, refilling time, blowing time etc. and the impact of these parameters on the process output

KU9. Various types of plastics like thermoplastics/ thermosetting plastics and the additives to be used

KU10. Properties of various plastic materials

KU11. Processing behavior of various plastic raw materials

KU12. Safe storage of raw materials

KU13. Different types of moulds & dies

KU14. Different types of measuring instruments like vernier callipers, micrometres etc

KU15. Geometry and dimension measurement of the product

KU16. Different types of tools to trim the plastic product

KU17. Hazards and safety aspects involved in different processing techniques

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Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Prepare document related to processing parameter, other technical records like machine log sheets, job card etc.
- GS2.** Prepare draft drawings for the final output product
- GS3.** Write information documents to internal departments/ internal teams
- GS4.** Read & interpret machine parameters
- GS5.** Read and interpret engineering drawing and sketches
- GS6.** Read equipment manuals and process documents
- GS7.** Read instructions like safety instructions , symbols while using the equipment in the plant area
- GS8.** Communicate orally any instructions related to work with superiors & co workers with clarity
- GS9.** Listen actively
- GS10.** Follow company protocol for communication
- GS11.** Make proper decisions pertaining to the work
- GS12.** Identification of problem
- GS13.** Finding the resource to resolve the problem
- GS14.** Consult superiors in case of any assistance
- GS15.** Plan, fix up priorities for work operations as per job requirements
- GS16.** Organize and analyze information relevant to work
- GS17.** Basic concepts of shop-floor work productivity including material management waste reduction etc.
- GS18.** undertake and express new ideas and initiatives to others
- GS19.** modify work plan to overcome unforeseen difficulties or developments that occur as work progresses
- GS20.** participate in improvement procedures including process, quality etc
- GS21.** apply, analyze, and evaluate the information gathered from observation, experience, reasoning, or communication, as a guide to thought and action
- GS22.** exhibit good team work with all
- GS23.** Consult superiors or fellow workers in case of any assistance
- GS24.** Maintains good inter personal relations

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Understand the work order and the process requirements</i>	6	12	-	-
PC1. Discuss the work order (work output) required from the process and with the supervisor	1	3	-	-
PC2. Refer all components / process related documents to understand dimensions and properties of the required work output	1	3	-	-
PC3. Understand the process requirements in terms of temperature of the heater, hydraulic pressure/ air pressure/ vacuum pressure, rotating speed of the screw pressure, injection time, refilling time, blowing time etc. as mentioned in the Work Instruction/ SOP/ Control Diagrams	2	3	-	-
PC4. Clearly understanding the does and don'ts of the manufacturing process as defined in SOPs/ Work Instructions or defined by supervisors	2	3	-	-
<i>Arrange for the material to be processed and apparatus required for the same</i>	9	17	-	-
PC5. Understand the conversion procedure and process to be adopted for completing the work order from the supervisor by referring the Work Instruction document/ SOP manual	2	3	-	-
PC6. Set the various parameters like temperature of the heaters, hydraulic pressure/air pressure/ vacuum pressure, rotating speed of the screw, screw pressure, regulating current, flow of coolant/ water etc. before starting the process as per the parameters are mentioned in the Work Instructions/ SOP manual	2	3	-	-
PC7. Understand the raw material like plastics granules, bonding additives etc. required for executing the activity	2	3	-	-
PC8. Ensure that the required material is available before starting the process	1	2	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC9. Understand the type of Mould /Die required for executing the required conversion operation and ensure that the same is available for moulding operations	1	3	-	-
PC10. Ensure the availability of spare parts for continuous operation of machine	1	3	-	-
<i>Clean the apparatus and the components before executing the process</i>	3	9	-	-
PC11. Ensure that mould / Die are cleaned properly & no foreign material is entrapped in parts of mould/die	1	3	-	-
PC12. Ensure cleaning of the other moulding machine tools, auxiliaries(if any)	1	3	-	-
PC13. Ensure cleaning of the area around the machine for any oil, grease, water etc	1	3	-	-
<i>Escalations of queries on the given job</i>	2.5	3.5	-	-
PC14. Consult with superiors in case of any doubt/clarification	1	1	-	-
PC15. Self-confidence after resolving the queries to complete the task	1	1	-	-
PC16. Report completion of work to superiors	0.5	1.5	-	-
<i>Interaction with other concern department</i>	1.5	4.5	-	-
PC17. Good interpersonal relations with superiors & fellow operators	0.5	1.5	-	-
PC18. Disciplined behavior in work place	0.5	1.5	-	-
PC19. Good coordination with other department person for getting their support for work	0.5	1.5	-	-
NOS Total	22	46	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	CPC/CAP/N0101
NOS Name	Basic concepts, job requirements & basic related process
Sector	Chemical & PetroChemical
Sub-Sector	
Occupation	Plastics Processing, Plastics Processing
NSQF Level	3.5
Credits	1
Version	1.0
Last Reviewed Date	26/05/2025
Next Review Date	25/05/2028
NSQC Clearance Date	26/05/2025

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Assessment System Overview:

- Batches are assigned to Training Assessment Wing (TAW), CIPET HO for planning of assessment
- Training Centers request TAW for Assessment and Certification of Trainees
- TAW identifies suitable assessor and nominates the assessor to the respective Training Centre
- TAW monitors the assessment process
- Training Centers maintain necessary records

2. Testing Environment:

- Check the Assessment location, date and time
- If the batch size is more than 30, then there should be 02 Assessors in a day (or) 01 Assessor in 2 days

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- Check that the allotted time to the candidates to complete the Theory & Practical Assessment

3. Assessment Quality Assurance levels/Framework:

- Question bank / Question Paper is prepared by the Subject Matter Experts (SME) / Assessor
- Questions are mapped to the specified assessment criteria
- Certified Assessor & Trainer will be engaged in the process

4. Types of evidence or evidence-gathering protocol:

- Date / Time recorded for the reporting of the assessor from assessment location
- Assessment batch - Group Photo of Trainees along with Assessor

5. Method of verification or validation:

- Surprise visit to the assessment location
- Virtual meet with the Assessor / Trainees

6. Method for assessment documentation, archiving, and access

- Hard copies of the documents are stored, soft copies of assessment evidences are stored in Email for future correspondence

Minimum Aggregate Passing % at QP Level : 50

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
DGT/VSQ/N0101.Employability Skills (30 Hours)	20	30	-	-	50	10
CPC/CAP/N0402.Basics of computer and data entry in MS OFFICE/office Open source suite Software	8	14	0	0	22	10

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National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
CPC/CAP/N0401.Basic health and safety practices at the workplace, 5S	10	30	0	0	40	10
CPC/CAP/N0105.Familiarized with various Plastics processing techniques & to assist the Operator in Extrusion & its Troubleshooting	50	100	0	0	150	15
CPC/CAP/N0104.Familiarized with various Plastics processing techniques & to assist the Operator in Blow Moulding & its Trouble shooting etc	30	70	0	0	100	15
CPC/CAP/N0103.Familiarized with various Plastics processing techniques & to assist the Operator in Injection Moulding & its Troubleshooting	50	100	0	0	150	15
CPC/CAP/N0102.To know about different plastic material	20	30	0	0	50	15
CPC/CAP/N0101.Basic concepts, job requirements & basic related process	22	46	0	0	68	10
Total	210	420	-	-	630	100



Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training

Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

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Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.